

The exponential distribution of the order of demonstrative, numeral, adjective and noun.

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ABSTRACT

The frequency of the preferred order for a noun phrase formed by demonstrative, numeral, adjective and noun has received significant attention over the last two decades. We investigate the actual distribution of the 24 possible orders. There is no consensus on whether it is well-fitted by an exponential or a power law distribution. We find that an exponential distribution is a much better model. This finding and other circumstances where an exponential-like distribution is found challenge the view that power-law distributions, e.g., Zipf’s law for word frequencies, are inevitable. We also investigate which of two exponential distributions gives a better fit: an exponential model where the 24 orders have non-zero probability (a geometric distribution truncated at rank 24) or an exponential model where the number of orders that can have non-zero probability is variable (a right-truncated geometric distribution). When consistency and generalizability are prioritized, we find higher support for the exponential model where all 24 orders have non-zero probability. These findings strongly suggest that there is no hard constraint on word order variation and then unattested orders merely result from undersampling, consistently with Cysouw’s view.

KEYWORDS

noun phrase; exponential distribution; power law distribution; word order

1. Introduction

The frequency of the preferred order of a noun phrase formed by demonstrative (D), numeral (N), adjective (A) and noun (n) has received substantial attention over the last 20 years (Cinque, 2005; Cysouw, 2010; Dryer, 2018; Martin, Holtz, Abels, Adger, & Culbertson, 2020; Medeiros, 2018). The 24 possible orders are listed in the 1st column of Table 1. Researchers have attempted to shed light on the actual variation of frequency among the 24 possible orders with some degree of precision (Cinque, 2005; Cysouw, 2010; Dryer, 2018). A key research question is why not all possible orders are attested. Certain researchers have attributed this to the existence of hard constraints limiting word order variation (Cinque, 2005, 2013; Medeiros, 2018; Medeiros, Piattelli-Palmarini, & Bever, 2016). A hard constraint is a constraint that makes certain word orders impossible. A soft constraint is one that reduces the probability of certain

orders, but never to zero probability. In contrast, Cysouw hypothesized that all orders are *a priori* possible but some are not attested due to undersampling (Cysouw, 2010).

Concerning the distribution of the rank r of a preferred order (the most frequent order has rank 1, the second most preferred order has rank 2, and so on), shown in Figure 1 top, Cysouw (2010) proposed an exponential distribution while Martin et al. (2020) assumed a power law distribution. Therefore, another key question is which of the two distributions is more appropriate. The frequency of rank r , i.e. $f(r)$, would follow a power law distribution if $f(r)$ could be approximated by

$$f(r) = cr^{-\alpha}, \quad (1)$$

where c and α are some positive constants. The parameter α is the so-called exponent of the power law. $f(r)$ would follow an exponential distribution if it could be approximated by

$$f(r) = ce^{-\beta r}, \quad (2)$$

where c and β are some positive constants.¹

In the remainder of the article, we address the two key questions above. We will show that an exponential distribution is a much better model than a power law distribution and will discuss its implications for the existence of hard constraints in the noun phrase and the meaningfulness of linguistic laws.

2. Methods

2.1. Data

Table 1 shows the frequency of the preferred order of demonstrative, numeral, adjective and noun in our dataset. We borrow these frequencies from two datasets: Dryer (2006) and Dryer (2018). Cysouw (2010, Appendix) displays the frequencies in Dryer (2006). In Dryer (2018), the frequency of each preferred order is available in languages, genera and adjusted number of languages. A genus (*genera* in plural) is a notion introduced by Dryer (1989) to refer to a group of closely related languages that is, essentially, an intermediate genetic classification between a language family and a language. Such a classification was applied notably in the World Atlas of Linguistic Structures.² The adjusted number of languages is an adjustment to the number of languages introduced by Dryer (2018) to control for geographic proximity or genetic relatedness. Frequencies in languages or genera are integer numbers while the frequency in adjusted number of languages is non-integer.

The frequencies above differ from frequencies used in traditional corpus studies in the sense that they are calculated on the preferred orders of languages. Consider the

¹Notice that $\alpha, \beta \geq 0$ by the definition of r .

²<https://wals.info/languoid/genealogy> A genus represents languages that are clearly related, but not so distantly that their relationship is controversial. It typically corresponds to a time depth of about 3,000–4,000 years of divergence, hence it is deeper than subfamilies (like Romance) but shallower than broad families (like Indo-European). For instance,

- Indo-European family $\rightarrow$ Romance genus $\rightarrow$ Catalan, French, Italian, ...
- Niger-Congo family $\rightarrow$ Bantu genus $\rightarrow$ Swahili, Zulu, Xhosa, ...

Languages that are genetically or geographically close may not be statistically independent (Winter & Grice, 2021).

Table 1. The frequency of the preferred order of the 24 orders of noun (n), adjective (A), numeral (N) and demonstrative (D) according to Dryer (2018) and Dryer (2006). Orders are sorted according to their frequency in languages.

order	Dryer (2018)			Dryer (2006)
	languages	genera	adjusted languages	frequency
nAND	182	85	44.17	88
DNA _n	113	57	35.56	58
N _n AD	67	27	14.54	32
D _n AN	53	40	29.95	19
DN _n A	40	32	22.12	18
nADN	36	19	14.8	19
nDAN	13	11	9	8
D _n NA	12	10	9.75	1
DA _n N	12	7	5.34	12
nNAD	11	9	9	6
nDNA	8	6	5.67	4
NA _n D	8	5	4	2
A _n ND	5	3	3	1
N _n DA	5	3	3	4
A _n DN	5	3	2.5	2
DAN _n	3	2	2	0
NDAN	2	2	2	0
nNDA	1	1	1	1
NAD _n	0	0	0	0
ND _n A	0	0	0	1
AD _n N	0	0	0	0
ADN _n	0	0	0	0
AND _n	0	0	0	0
AN _n D	0	0	0	0

frequency of nAND in languages, that is 182 according to Dryer (2018) (Table 1). This frequency is the number of languages that prefer the order nAND (and not any of the other 23 possibilities) for a noun phrase consisting of a demonstrative, a numeral, an adjective and a noun. It is not the frequency of that order in a corpus, although one would expect nAND to be the most frequent in a corpus in each of the 182 languages.

For convenience, we define F_x as

$$F_x = \sum_{r=1}^{r_{max}} f(r)r^x,$$

where r_{max} is the maximum rank in the sample. Thus, r_{max} is also the number of attested orders, i.e. the number of distinct orders observed in the sample. The quantity F_0 is just the total frequency and is also the sample size. The quantity F_1 is the sum of the ranks in our sample (every rank contributes to the sum with as many summands as its frequency). The average rank is then

$$\langle r \rangle = F_1/F_0.$$

Table 2. Summary of the elementary statistical properties of each dataset: the total frequency (F_0), the average frequency rank ($\langle r \rangle$) and the maximum frequency rank in the sample (r_{max}).

dataset	unit	F_0	$\langle r \rangle$	r_{max}
2006	languages	276	3.5	17
2018	languages	576	3.7	18
	genera	322	4.2	18
	adjusted languages	217.4	4.8	18

Table 2 summarizes the elementary statistical properties of the datasets. Within the 2018 dataset, the sample size (F_0) reduces while the average rank increases as one moves from languages, to genera and then to adjusted number of languages.

2.2. The models

The term power law is used both for continuous and discrete random variables (Conrad & Mitzenmacher, 2004; Debowski, 2020; Naranan & Balasubrahmanyam, 1998; Newman, 2005; Stumpf & Porter, 2012). Similarly, the term exponential distribution is used both for continuous and discrete random variables (Broido & Clauset, 2019; Clauset, Shalizi, & Newman, 2009; Ferrer-i-Cancho, 2004; Newman, 2005). In these contexts, the term power law is used to refer to a distribution that approximates a power-law function. This is why the term power-law-like distribution is often used. Similarly, the term exponential is used to refer to a distribution that approximates an exponential function. Analogously, one can also use the term exponential-like distribution. However, here our random variable, i.e. rank, is discrete. Martin et al. (2020) stated that the distribution is a power law but did not specify the actual form of the distribution. Our first task is to translate informal terminology into specific discrete distributions (Johnson, Kemp, & Kotz, 2005; Wimmer & Altmann, 1999).

As there cannot be more than $N = 24$ orders, here we are interested in right-truncated models for $p(r)$, namely models that give $p(r) = 0$ for $r > N$. Among these models, we are interested in models with early right-truncation, namely models that have a parameter $R \leq N$ such that $p(r) = 0$ when $r < 1$ or $r > R$ and $p(r) > 0$ for $1 \leq r \leq R$. A power-law-like distribution (Equation 1) can be specified as a zeta distribution, namely (Wimmer & Altmann, 1999, 664-665)

$$p(r) = cr^{-\alpha}, \quad (3)$$

where the normalization factor is $c = 1/\zeta(\alpha)$. In turn, $\zeta(\alpha)$ is the Riemann zeta function, i.e.

$$\zeta(\alpha) = \sum_{r=1}^{\infty} r^{-\alpha}. \quad (4)$$

A zeta distribution is not an adequate power-law model for our setting because that model predicts $p(r) > 0$ for any finite r such that $r > N$. For this reason, we consider instead a right-truncated zeta distribution with two parameters, α and R , such that

the normalization factor becomes (Wimmer & Altmann, 1999, 577-578)

$$c = \frac{1}{H(\alpha, R)}. \quad (5)$$

In turn, $H(R, \alpha)$ is the generalized harmonic number in power α i.e.

$$H(\alpha, R) = \sum_{r=1}^R r^{-\alpha}.$$

An exponential-like distribution (Equation 2) can be specified as a geometric distribution, i.e.

$$p(r) = c(1 - q)^{r-1}, \quad (6)$$

where c is a normalization factor and $q \in (0, 1)$ is a parameter. The untruncated geometric distribution is obtained with $c = q$ and then q is the only parameter. An adequate version in our setting is the 2-parameter right-truncated geometric distribution, where (Appendix B)

$$c = \frac{q}{1 - (1 - q)^R}. \quad (7)$$

Technically, the geometric distribution is the discrete analog of the exponential distribution, which is usually defined on continuous random variables (Johnson et al., 2005, p. 210) (see Appendix A for the relationship between Equation 2 and 6).

In this article, we use the following ensemble of models (the nickname of the model is followed by its definition):

- *Zeta 2*. The 2-parameter right-truncated zeta distribution (Equation 3 with c defined by Equation 5). The two parameters are α and R .
- *Zeta 1*. The 1-parameter right-truncated zeta distribution that is obtained by setting $R = N$ in the Zeta 2 model. The only parameter is α .
- *Geometric 2*. The 2-parameter right-truncated geometric (Equation 6 with c defined by Equation 7). The two parameters are q and R .
- *Geometric 1*. A 1-parameter right-truncated geometric that is obtained by setting $R = N$ in the Geometric 2 model. The only parameter is q .

Table 3 summarizes the mathematical definition of each model and its parameters. Recall that we have excluded from the ensemble popular 1-parameter models such as the (untruncated) geometric model or the zeta distribution because r cannot be larger than $N = 24$.

2.3. Visual diagnostic

Consider a plot with $p(r)$ on the y -axis and r on the x -axis (Figure 1). A preliminary conclusion about the functional dependence between $p(r)$ and r can be obtained by taking logarithms on one of the axes or both. If r followed a power law (Equation 3), $\log p(r)$ would be a linear function of $\log r$ since

$$\log p(r) = -\alpha \log r + \log c.$$

Table 3. The ensemble of models. For each model, we show its definition, the free parameters and the theoretical constraints on the parameters.

model	definition	parameters	constraints
Zeta 1	$p(r) = \begin{cases} \frac{1}{H(N,\alpha)} r^{-\alpha} & \text{if } r \in [1, N] \\ 0 & \text{if } r \notin [1, N] \end{cases}$	α	$0 \leq \alpha$
Zeta 2	$p(r) = \begin{cases} \frac{1}{H(R,\alpha)} r^{-\alpha} & \text{if } r \in [1, R] \\ 0 & \text{if } r \notin [1, R] \end{cases}$	α, R	$0 \leq \alpha, 1 \leq R \leq N$
Geometric 1	$p(r) = \begin{cases} \frac{q}{1-(1-q)^N} (1-q)^r & \text{if } r \in [1, N] \\ 0 & \text{if } r \notin [1, N] \end{cases}$	q	$0 < q < 1$
Geometric 2	$p(r) = \begin{cases} \frac{q}{1-(1-q)^R} (1-q)^r & \text{if } r \in [1, R] \\ 0 & \text{if } r \notin [1, R] \end{cases}$	q, R	$0 < q < 1, 1 \leq R \leq N$

Then the plot in double logarithmic scale should show a straight line with a negative slope $-\alpha$. If r followed a geometric distribution (Equation 6), $\log p(r)$ would be a linear function of r since

$$\log p(r) = (r - 1) \log(1 - q) + \log c.$$

Then the plot in linear scale for the x -axis and logarithmic scale for the y axis should show a straight line with a negative slope $\log(1 - q)$.

2.4. Model selection

We use information-theoretic model selection to find the best model in the ensemble. We use the corrected Akaike Information criterion (AIC_c) and the Bayesian Information Criterion (BIC), that are defined as (Burnham & Anderson, 2002; Wagenmakers & Farrell, 2004)

$$\begin{aligned} AIC_c &= -2\mathcal{L} + 2K \frac{F_0}{F_0 - K - 1} \\ BIC &= -2\mathcal{L} + K \log F_0, \end{aligned} \tag{8}$$

where $\mathcal{L}$ is the maximum log-likelihood of the parameters of the model, and K is the number of parameters of the model. The best model is the one that minimizes a criterion.

We define $\Delta_i(x)$ as the difference in a score x between the i -th model and the best model. Thus, $\Delta_i(BIC)$ is the difference in BIC between the i -th model and the best model.

We define $w_i(x)$, the weight of model i according to a score x , as

$$w_i(x) = \frac{\exp\left(-\frac{1}{2}\Delta_i(x)\right)}{\sum_j \exp\left(-\frac{1}{2}\Delta_j(x)\right)}.$$

$w_i(AIC_c)$, the AIC weight, estimates the probability that model i is the true model of the ensemble (Burnham & Anderson, 2002; Wagenmakers & Farrell, 2004). $w_i(BIC)$, the BIC weight, estimates the probability that model i is the quasi-true model in the

ensemble (Burnham & Anderson, 2002, p. 297). The evidence of model i over model j with respect to some score x is defined as the ratio $\frac{w_i(x)}{w_j(x)}$. For AIC and BIC , one has (Wagenmakers & Farrell, 2004)

$$\begin{aligned}\frac{w_i(AIC)}{w_j(AIC)} &= \frac{L_i}{L_j} \exp(K_j - K_i) \\ \frac{w_i(BIC)}{w_j(BIC)} &= \frac{L_i}{L_j} n^{\frac{1}{2}(K_j - K_i)},\end{aligned}$$

where L_i , K_i , are the likelihood and the number of parameters of model i . For AIC_c , it is easy to see that

$$\frac{w_i(AIC_c)}{w_j(AIC_c)} = \frac{L_i}{L_j} \exp \left[F_0 \left(\frac{K_j}{F_0 - K_j - 1} - \frac{K_i}{F_0 - K_i - 1} \right) \right].$$

An obvious difference between AIC_c and BIC is that BIC introduces a stronger penalty for lack of parsimony than AIC_c (Wagenmakers & Farrell, 2004). BIC is more useful in selecting a correct model, which is our primary aim, while AIC is more appropriate in finding the best model for predicting future observations (Chakrabarti & Ghosh, 2011). AIC is not a consistent score in the sense that as the number of observations F_0 tends to infinity, the probability that AIC recovers a true low-dimensional model does not converge to 1 (Wagenmakers & Farrell, 2004).

We revisit the derivation of $\mathcal{L}$ for the Zeta 2 model (Baixeries, Elvevåg, & Ferrer-i-Cancho, 2013) and extend it to the Geometric 2 model. The likelihood of a sample of ranks $\{r_1, \dots, r_i, \dots, r_{F_0}\}$ can be expressed as

$$L = \prod_{i=1}^{F_0} p(r_i) \quad (9)$$

and then $\mathcal{L} = \log L$ can be expressed as

$$\mathcal{L} = \sum_{r=1}^{r_{max}} f(r) \log p(r). \quad (10)$$

For the Zeta 2 model (Equation 3 with Equation 5), the log-likelihood is

$$\mathcal{L} = -\alpha \sum_{r=1}^{r_{max}} f(r) \log r - F_0 \log H(R, \alpha). \quad (11)$$

For the Geometric 2 model, we derive $\mathcal{L}$ by plugging Equation 6 (with c defined as in Equation 7) into the general definition of log-likelihood in Equation 10. After some algebra, one obtains

$$\mathcal{L} = F_0 \log \frac{q}{1 - (1 - q)^R} + (F_1 - F_0) \log(1 - q). \quad (12)$$

To obtain the best parameters of a model by maximum likelihood, we proceed as follows. If the model has one parameter, we use one-dimensional optimization. For

the Geometric 1 model, the parameter q is optimized in the interval $(0, 1)$. For the Zeta 1 model, the parameter α is optimized in the interval $[0, 10^6]$. For the models that have two parameters, we note that maximum $\mathcal{L}$ requires $R = r_{max}$ (Appendix C). Therefore, to maximise $\mathcal{L}$, we set parameter R to r_{max} and optimize the other parameter following the same procedure as for the 1-parameter models.

3. Results

3.1. Visual diagnostic

We examine the look of plots of $p(r)$ as a function of r (Figure 1) following the reasoning in Section 2.3. In normal scale, a decreasing curve is observed (Figure 1 top) and it is difficult to determine whether the distribution is power-law-like or exponential-like. In logarithmic scale only for $p(r)$, a straight line with negative slope appears, which is compatible with a geometric distribution (Figure 1 middle). In double logarithmic scale, $p(r)$ curves down, which is incompatible with a power law function (Figure 1 bottom). If a distribution curves down in double logarithmic scale, that implies that the probability decay is faster than expected for a power law distribution. To sum up, an exponential distribution is a better candidate than a power law.

3.2. Model selection

Table 4. Summary of the best parameters. For every dataset, frequency unit and model, we show the value of the parameters that maximize $\mathcal{L}$ (R is the number of non-zero probability ranks, α is the exponent of the power-law models and q is a parameter of the geometric models).

dataset	unit	model	R	α	q
2006	languages	Geometric 1			0.282
		Geometric 2	17		0.277
		Zeta 1		1.386	
		Zeta 2	17	1.271	
2018	languages	Geometric 1			0.269
		Geometric 2	18		0.265
		Zeta 1		1.361	
		Zeta 2	18	1.264	
	genera	Geometric 1			0.238
		Geometric 2	18		0.231
		Zeta 1		1.237	
		Zeta 2	18	1.126	
	adjusted languages	Geometric 1			0.204
		Geometric 2	18		0.193
		Zeta 1		1.095	
		Zeta 2	18	0.963	

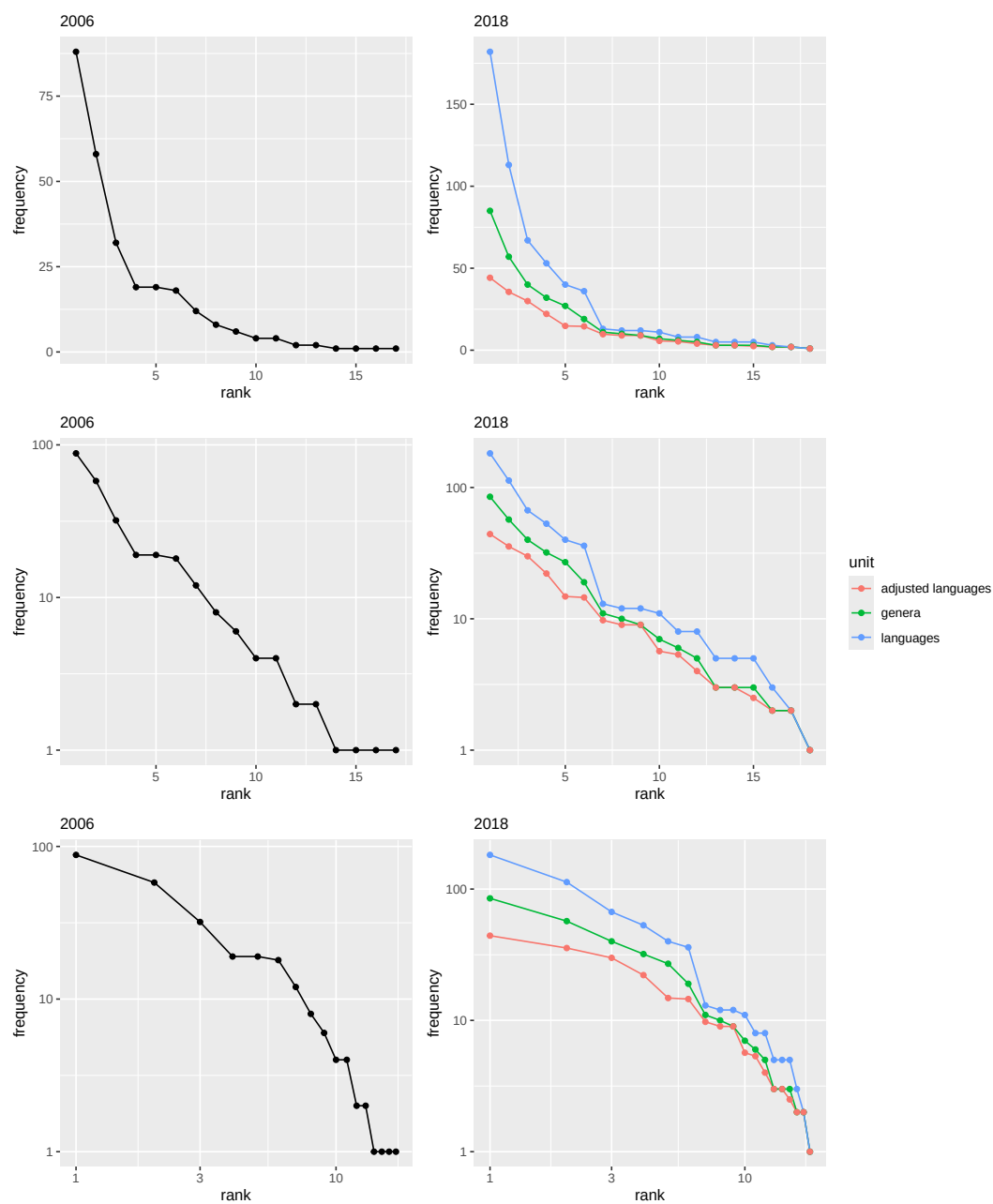


Figure 1. The frequency of a word order of rank r , $f(r)$, in three distinct scales: normal (top), linear-log (middle) and log-log (bottom). Left. Frequency is measured in languages according to Dryer (2006). Right. Frequency is measured in languages, genera and adjusted number of languages according to Dryer (2018).

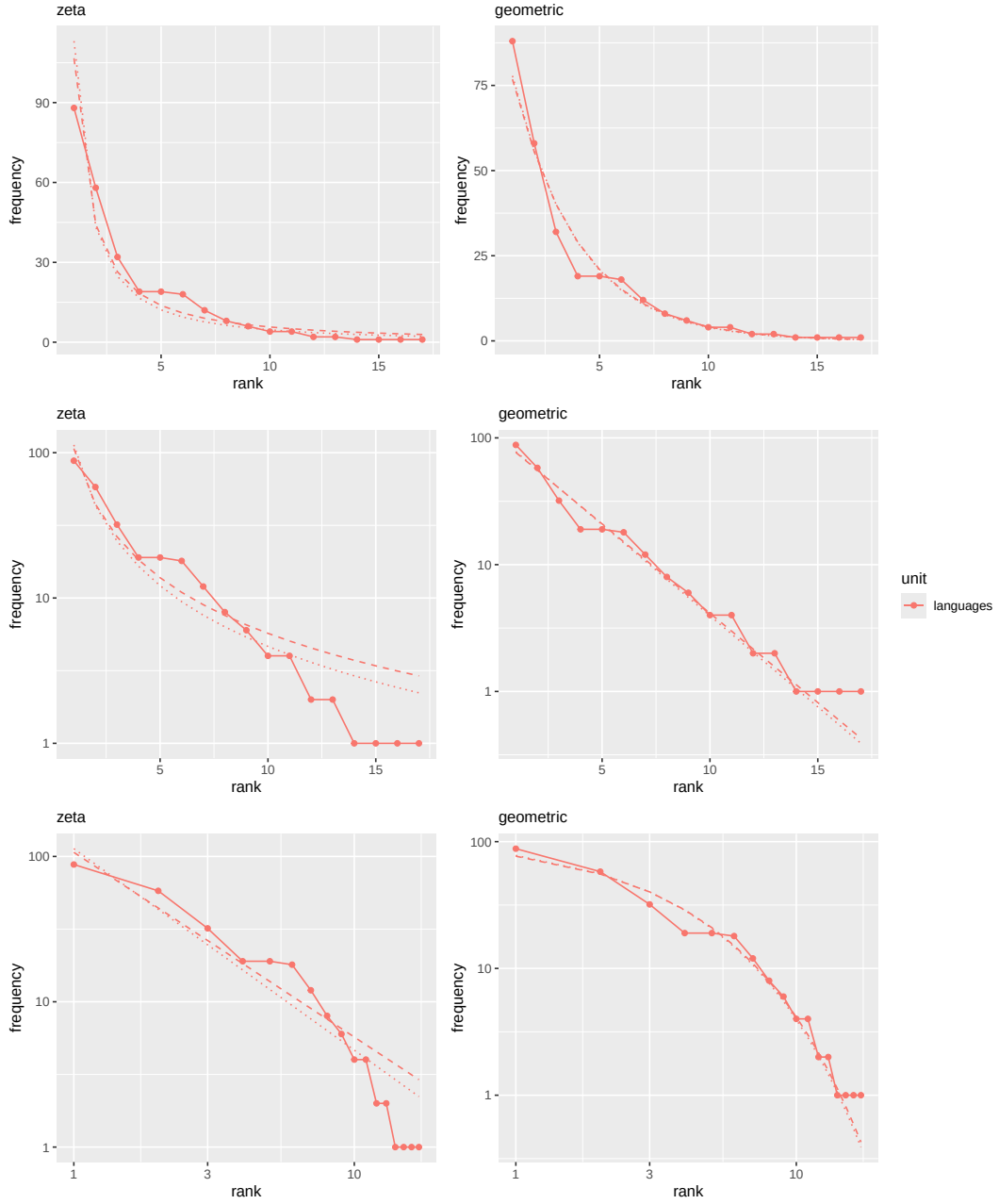


Figure 2. $f(r)$, the frequency of a word order of rank r in Dryer (2006), in three distinct scales: normal (top), linear-log (middle) and log-log (bottom). The solid line is the real curve and the discontinuous lines are the expected value of $f(r)$, that is $\mathbb{E}[f(r)] = F_0 p(r)$, where $p(r)$ is defined by the best fit of a model (Table 4). Left. Real curves versus the best fit of the Zeta 1 model (dotted line) and that of Zeta 2 model (dashed line). Right. Real curves versus the best fit of Geometric 1 (dotted line) and that of Geometric 2 (dashed line).

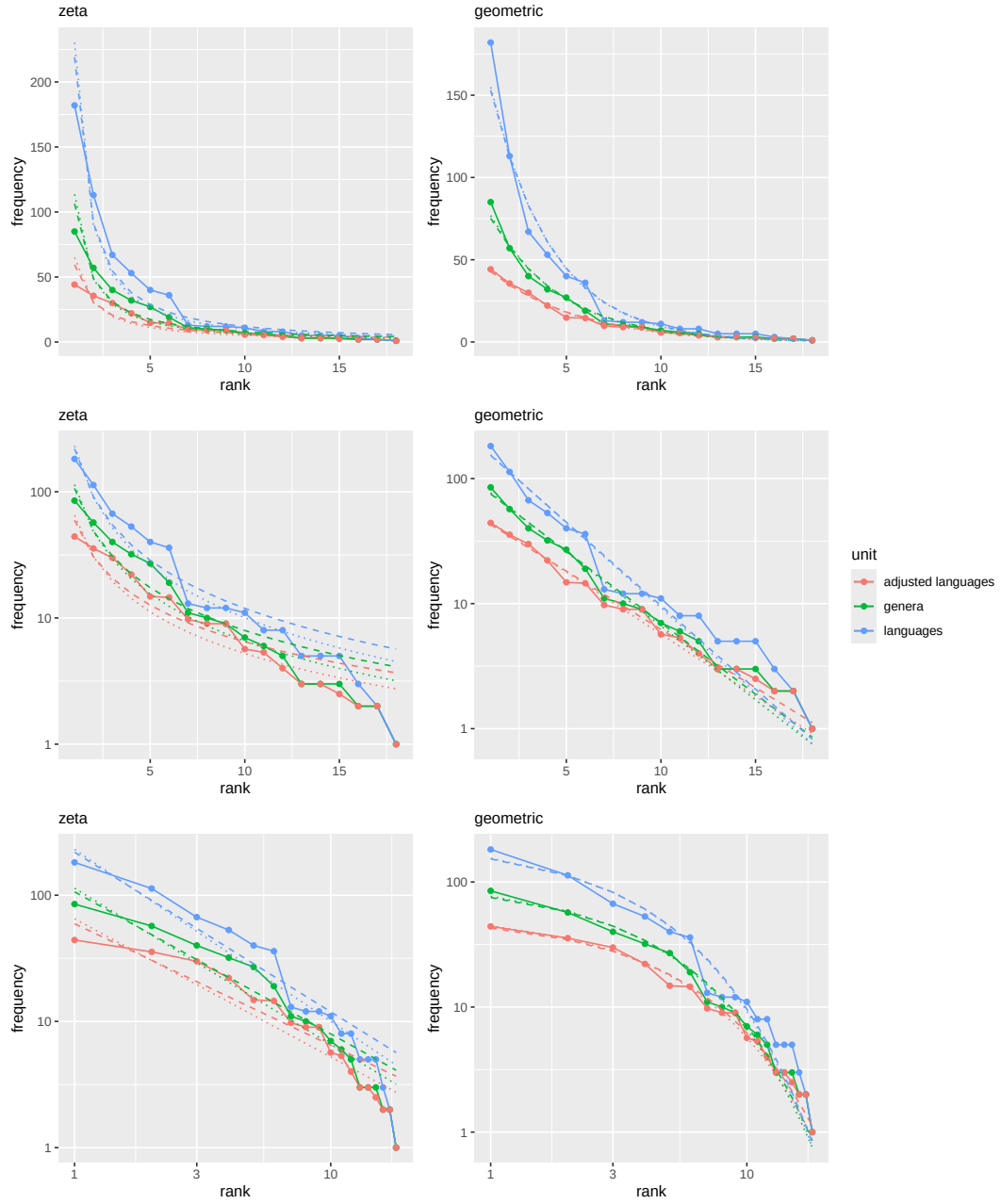


Figure 3. $f(r)$, the frequency of a word order of rank r in Dryer (2018). The format is the same as in Figure 2.

Table 5. Summary of the model selection. For every dataset, frequency unit and model, we show the log-likelihood ($\mathcal{L}$), the corrected Akaike Information Criterion (AIC_c), the AIC_c difference ($\Delta(AIC_c)$), the AIC_c weight ($w(AIC_c)$), the Bayesian Information Criterion (BIC), the BIC difference ($\Delta(BIC)$) and the BIC weight ($w(BIC)$).

dataset	unit	model	$\mathcal{L}$	AIC_c	$\Delta(AIC_c)$	$w(AIC_c)$	BIC	$\Delta(BIC)$	$w(BIC)$
2006	languages	Geometric 1	-581.5	1165	0	0.518	1168.6	0	0.866
		Geometric 2	-580.6	1165.2	0.1	0.482	1172.4	3.7	0.134
		Zeta 1	-603.3	1208.7	43.7	$1.7 \cdot 10^{-10}$	1212.3	43.7	$2.9 \cdot 10^{-10}$
		Zeta 2	-589.8	1183.6	18.6	$4.8 \cdot 10^{-5}$	1190.8	22.2	$1.3 \cdot 10^{-5}$
2018	languages	Geometric 1	-1244.8	2491.6	1.7	0.304	2496	0	0.793
		Geometric 2	-1243	2490	0	0.696	2498.7	2.7	0.207
		Zeta 1	-1278.7	2559.4	69.4	$5.9 \cdot 10^{-16}$	2563.7	67.7	$1.5 \cdot 10^{-15}$
		Zeta 2	-1254.5	2513.1	23.1	$6.7 \cdot 10^{-6}$	2521.8	25.8	$2 \cdot 10^{-6}$
	genera	Geometric 1	-738.5	1478.9	2.1	0.255	1482.7	0	0.691
		Geometric 2	-736.4	1476.8	0	0.745	1484.3	1.6	0.309
		Zeta 1	-766.3	1534.5	57.8	$2.1 \cdot 10^{-13}$	1538.3	55.6	$5.8 \cdot 10^{-13}$
		Zeta 2	-748.6	1501.3	24.6	$3.5 \cdot 10^{-6}$	1508.8	26.2	$1.4 \cdot 10^{-6}$
	adjusted languages	Geometric 1	-532.8	1067.7	3.9	0.123	1071	0.6	0.427
		Geometric 2	-529.8	1063.7	0	0.877	1070.4	0	0.573
		Zeta 1	-555.6	1113.3	49.5	$1.5 \cdot 10^{-11}$	1116.6	46.2	$5.4 \cdot 10^{-11}$
		Zeta 2	-539.7	1083.5	19.8	$4.4 \cdot 10^{-5}$	1090.2	19.8	$2.9 \cdot 10^{-5}$

A stronger conclusion on the best model is obtained by inspecting the AIC/BIC scores (Table 5). First, the geometric distribution models give lower AIC_c or BIC than the power law distribution models. The AIC_c weights indicate that the power law model is unlikely to be the true model of the ensemble ($w < 5 \cdot 10^{-5}$ for power law models). Similarly, the BIC weights indicate that the power law model is unlikely to be the quasi-true model of the ensemble ($w < 3 \cdot 10^{-5}$ for power law models). Visual inspection confirms it: the theoretical curve of the geometric model is closer to the actual distribution than the theoretical model of the power law (Figures 2 and 3). The power-law fails simultaneously by overestimating $p(1)$ and not capturing the faster decay of the actual distribution. The best values of q_i , the value of q of Geometric i model, indicate that $|q_1 - q_2|$ is small but always $q_1 > q_2$ (Table 4). In addition, q_i is close to $1/\langle r \rangle$, the maximum likelihood estimator of q for an untruncated geometric distribution (Equation 6 with $c = q$).

When comparing Geometric 1 and Geometric 2 with information criteria, the results depend on the score. AIC_c provides more support for Geometric 2 while BIC provides more support for Geometric 1. In particular,

- AIC_c . Geometric 2 is better than Geometric 1 in terms of AIC_c , except in the 2006 dataset, where the AIC of Geometric 2 is just 0.1 nats above Geometric 1. We consider the evidence of Geometric 2 over Geometric 1 by means of $w_2(AIC_c)/w_1(AIC_c)$. In the 2006 dataset, Geometric 2 is about as likely to be the true model as Geometric 1 ($w_2(AIC_c)/w_1(AIC_c) \approx 1$). In the 2018 dataset, Geometric 2 is about 2 times more likely to be the quasi-true model than Geometric 1 when frequency is measured in languages, about 3 times more likely when frequency is measured in genera and about 7 times higher when frequency is measured in adjusted number of languages.
- BIC . Geometric 1 is better than Geometric 2 in terms of BIC , except for adjusted number languages in the 2018 dataset, where the BIC of Geometric 1 is just 0.6 nats above Geometric 2. However, the sample size is the smallest for adjusted number of languages (Table 2) which is a hindrance for model selection with BIC (Burnham & Anderson, 2002, p. 288); recall the dependence of BIC on sample size in Equation 8. We consider the evidence of Geometric 1 over Geometric 2 by means of $w_1(BIC)/w_2(BIC)$. The BIC weights indicate that Geometric 1 is about 6 times more likely to be the quasi-true model than Geometric 2 in the 2006 dataset ($w_1(BIC)/w_2(BIC) \approx 6$). In the 2018 dataset, Geometric 1 is about 4 times more likely to be the quasi-true model than Geometric 2 when frequency is measured in languages, about 7/3 times more likely when frequency is measured in genera but about 3/4 “higher” when frequency is measured in adjusted number of languages.

4. Discussion

4.1. Is the distribution of preferred orders in languages exponential or power-law?

4.1.1. The best distribution

We have shown that the geometric distribution, a discrete exponential-like distribution, is a much better model than a power law distribution, both according to visual diagnostic and model selection. Our conclusion is consistent with Cysouw’s proposal

(Cysouw, 2010) and at odds with Martin et al.’s assumption of a power law (Martin et al., 2020). The expectation of a power law has led to misclassify the rank distribution of vocalizations produced by other species as power laws by means of visual diagnostic (Dreher, 1961; Howes-Jones & Barlow, 1988; Janik, 2006). In Figure 12 (p. 24), Howes-Jones and Barlow (1988) show the frequency of the calls of the warbling vireo (*Vireo vilgus vilgus*) as a function of their rank. In figure 2 (p. 1800), Dreher (1961) shows the frequency of the “tonemes” produced primarily by dolphins (*Tursiops truncatus*). In both cases, authors conclude that finding a straight line in linear-log scale agrees with Zipf’s law for word frequencies. However, we have shown that a straight line in that scale is indeed indicative of an exponential distribution (Section 2.3). Thus power laws are less ubiquitous than usually believed.

4.1.2. The correct exponential distribution

We have found that the best model depends on the score. According to AIC_c , Geometric 2 has more evidence than Geometric 1 (except for the 2006 dataset). According to BIC , Geometric 1 has more evidence (except for adjusted number of languages in the 2018 dataset). However, a stronger conclusion can be reached by looking at the ability of the models to generalize in perspective. Cinque (2005) reported only 14 attested orders. Although he did not report the frequencies of each attested on a sample, we can be totally certain that the best fit of Geometric 2 by maximum likelihood would conclude that only $R = 14$ orders have non-zero probability (justification in Appendix C), which we know it would be wrong because 17 and 18 orders were found later on (Dryer, 2006, 2018). We also know that the best fit of Geometric 2 to the 2006 dataset, namely $R = 17$ is misleading because $R = 18$ for the 2018 dataset (Table 4). Thus, Geometric 2 fails to generalize twice.

When integrating likelihood into an information theoretic criterion and considering the 2006 dataset again, AIC_c weights conclude that Geometric 1 is as likely as Geometric 2 (recall $w_2(AIC_c)/w_1(AIC_c) \approx 1$) but BIC weights are able to realize that Geometric 1 is more likely to be the best (recall $w_1(BIC_c)/w_2(BIC_c) \approx 6$). Crucially, AIC_c does not foresee that the best model in the 2006 dataset, Geometric 2, produces a zero likelihood when applied to the 2018 dataset (notice that the application of the best fit of Geometric 2 for the 2006 dataset to the 2018 dataset yields $p(18) = 0$, which produces $L = 0$ when applied to equation 9). In contrast, the best model for the 2006 dataset according to BIC yields a non-zero likelihood when applied to the 2018 dataset. In sum, BIC catches early the model that generalizes while AIC_c is unable to realize that one of the models overfits the 2006 dataset. Therefore, AIC_c lacks generalizability.

The failure of AIC_c on the 2006 dataset is not surprising given the theoretical properties of AIC_c versus BIC (Wagenmakers & Farrell, 2004). AIC is not *consistent* in the sense that, as the number of observations (F_0 in our notation) grows very large, the probability that AIC_c recovers a true low-dimensional model does not converge to 1 (Bozdogan, 1987, p. 357). BIC is consistent as the number of observations tends to infinity. In our application, that means that if we supplied to AIC_c a dataset comprising all languages on Earth or even all languages that ever existed in our planet, it would not be warranted that AIC_c would recover the right geometric models while BIC would be much closer to finding the right geometric model. BIC is more useful in selecting a correct model (in this case, discarding an incorrect model) while the AIC is more appropriate in finding the best model for predicting future observations (Chakrabarti & Ghosh, 2011). In our application, the main goal is to find the right geo-

metric model, not a geometric model that predicts the actual shape of the distribution when future observations are incorporated.

Above, we have provided empirical evidence of the failure of AIC_c to find a correct model in our application. However, we wish to highlight the theoretical power of BIC for our research problem and our datasets. Once we have discarded the power-law models, the model selection reduces to choosing between two geometric models. BIC assumes equal prior probability on each model (Burnham & Anderson, 2002). That implies assigning initial equal chance to early truncation (Geometric 2) and to late truncation (Geometric 1). Indeed, such a contest between two nested models, Geometric 1 and Geometric 2, can be seen as a *dimensionality guessing problem* (Geometric 2 adds on dimension with respect to Geometric 1) and BIC is a consistent estimator of K , the dimensions of the “true model” (Burnham & Anderson, 2002, p. 284). BIC implicitly assumes that “*truth is of fairly low dimension (e.g., $K = 1 - 5$) and that the data-generating (true) model is fixed as sample size increases*” (Burnham & Anderson, 2002, 286). Various sources of evidence suggest that the true model’s dimensionality is bounded by a small number in our application. Dryer (2018) accurately modelled the frequency rank of the orders by means of 5 binary descriptive principles, suggesting that the true number of dimensions of the distribution is bounded above by $K = 5$. Indeed, just $\lfloor \log_2 N \rfloor + 1 = 5$ binary parameters suffice to sort N orders so as to match the desired frequency rank. Previously, Cysouw (2010) had proposed exponential models for the frequency of orders with 3, 4, or 6 binary parameters. Therefore, we apply BIC under favourable conditions, where the dimensions of truth are bounded. In our application, BIC always provides more evidence for Geometric 1 except for adjusted number of languages in the 2018 dataset, which coincides with the smallest sample (Table 2). The consistency property of BIC requires a large enough sample (Equation 8 and Burnham and Anderson (2002, p. 288)). Therefore, our findings so far and the conditions where we are applying BIC suggest that Geometric 1 is a stronger model for the underlying distribution.

4.2. Are there hard constraints on word order?

The quick answer to this question is that there is no statistically robust evidence for a hard constraint limiting word order variation in the noun phrase. A detailed discussion follows.

Once the power law models are discarded, the question of the existence of hard constraints reduces to a dimensionality guessing problem, i.e. which of the two geometric models is the best. The evidence for Geometric 1 is a challenge for the existence of hard constraints limiting word order variation in languages that would explain why not all 24 possible orders are attested (Cinque, 2005, 2013; Medeiros, 2018; Medeiros et al., 2016). If there were no hard constraints (only soft constraints), the best model should be Geometric 1; if strong constraints existed, Geometric 2 (with $R < 24$) should be the best model. The strength of Geometric 1 is consistent with Cysouw’s hypothesis: all orders are *a priori* possible but some are not attested due to undersampling (Cysouw, 2010). The challenge of proponents of a hard constraint is two-fold. First, to make a robust proposal, one that does not fall any time that new orders are attested. For instance, the proposal that the 14 orders attested by Cinque (2005) result from universal grammar or some universal cognitive mechanism, still stands as soft constraint but not as hard constraint (Cinque, 2005, 2013; Medeiros, 2018; Medeiros et al., 2016) because so far, 18 orders have been attested (Dryer, 2018). Model selection by means

of AIC_c commits overfitting for believing that the current number of attested orders is the true one. In particular, AIC_c leads to parameters for the best model on the data from Dryer (2006) that do not generalize to the data from Dryer (2018). Second, to deal with parsimony. Postulating a hard constraint implies a loss of parsimony but is it rewarding enough in terms of proximity to truth? Information criteria address this question and give a compelling answer: when empirical generalizability and theoretical consistency are prioritized (this is the virtue of BIC with respect to AIC_c), the absence of a hard constraint (Geometric 1) becomes more likely than its presence (Geometric 2), as shown in Table 5).

It is important to understand that the answer to the question on the existence of hard constraints does not follow from choosing BIC because it penalizes for lack of parsimony more strongly than AIC (Wagenmakers & Farrell, 2004). If we did so, the conclusion that Geometric 1, and thus the absence of hard constraints, would be trivial. The relevant questions, that summarize the discussions above, are the following:

- (1) Do we want a score for which there is no empirical evidence of contradiction in large enough samples?
- (2) Do we want a score that is theoretically consistent, namely, that as more languages are sampled, the probability that the score chooses the right low-dimensional model tends to one?
- (3) Do we want to use a score that is more useful in selecting a correct model (over being more appropriate in finding the best model for predicting future observations)?

If the answer to all the questions above is YES then

- The score is BIC .
- The best model is Geometric 1.
- The absence of a hard constraint is more likely than its presence.
- The use of score with a stronger penalty for lack of parsimony with respect to AIC is a side-effect, not a prior desideratum.

4.3. The meaningfulness of linguistic laws

The abundance of exponential-like distributions has implications for the debate on the meaningfulness of linguistic laws (Semple, Ferrer-i-Cancho, & Gustison, 2022, Box 2). Since Zipf’s foundational research (Zipf, 1949), many researchers have cast doubt on the depth and utility of linguistic laws such as Zipf’s rank-frequency law, the power law that characterizes the distribution of word ranks (Mehri & Jamaati, 2017; Moreno-Sánchez, Font-Clos, & Corral, 2016; Zipf, 1949). The recurrent criticism that linguistic laws in the form of power laws are inevitable (Miller & Chomsky, 1963; Solé, 2010) can be falsified by finding patterns across different species and systems that do not conform to the law that is claimed to be inevitable (Semple et al., 2022). Various sources of evidence demonstrate that Zipf’s rank-frequency law is not inevitable (Li, 2002). Here we review evidence from exponential distributions. First, our finding that an exponential distribution yields a better fit to word order ranks than a power-law. Second, the exponential distribution that is found in the order of SOV structures (Cysouw, 2010) as well as in part-of-speech tags (Tuzzi, Popescu, & Altmann, 2010), colors, kinship terms, verbal alternation classes (Ramscar, 2019). Third, the exponential rank distribution in the species mentioned above (Section 4.1.1) as well as in “key signs” produced by rhesus monkeys (Schleidt, 1973, Figure 3, p. 367). Fourth, the exponential

distribution is found in other linguistic variables such as the distance between syntactically related words (Ferrer-i-Cancho, 2004; Petrini & Ferrer-i-Cancho, 2025) or the length of vocal sequences in primates (Girard-Buttoz et al., 2022; Gustison, Semple, Ferrer-i-Cancho, & Bergman, 2016). Finally, in non-linguistic contexts, the projection distances between cortical areas exhibit an exponential distribution (Ercsey-Ravasz et al., 2013) and a double exponential distribution characterizes the average distance traversed by foraging ants (Campos, Bartumeus, Méndez, Andrade, & Espadaler, 2016), just to name a few. In sum, there is no empirical support for the claim that power laws are inevitable, even in a linguistic context

5. Conclusion

We have shown that a geometric distribution gives a better fit than a power law distribution to the distribution of preferred orders in the noun phase. This has implications for various components of the generative-linguistics program and other branches of theoretical linguistics that share the same abstractions, e.g., binary acceptability,³ or the same methods. On the one hand, our analysis shows that statistical evidence for a hard constraint on word order variation in the noun phrase is lacking. Not observing the 24 orders is not enough to claim a hard constraint. History has demonstrated that the post-hoc arguments built on the current number of attested orders (Cinque, 2005, 2013; Medeiros, 2018; Medeiros et al., 2016) have collapsed as more orders have been attested. On the other hand, the finding of an exponential distribution challenges another component of the generative-linguistics program, i.e. the inevitability of power laws such as Zipf’s rank-frequency law, and consequently, the lack of interest in explaining their origin (Miller & Chomsky, 1963). Interestingly, linguistic laws are absent from the agenda in the continuum between generative linguistics and its opponents. When claiming that linguistic universals are a myth, opponents ignore their existence or neglect them (Evans & Levinson, 2009). An intriguing question is whether they do it for the same reason as generativists and formal linguists from other traditions. While researchers engage in complex debates about the potential advantage of models with a huge number of parameters (Futrell & Mahowald, 2025) as if reality were intrinsically of ultra-high dimension, the examination of elementary exponential distributions with a few parameters reveals that a hard constraint on word order lacks statistical support, as we have seen here, while it sheds light on the structure of short-term memory and the dynamics of incremental sentence processing (Petrini & Ferrer-i-Cancho, 2025). Reality may be simple, but researchers may fail to see it.

APPENDIX

Appendix A. Exponential versus geometric distribution

The primary goal of this appendix is not to point out that the geometric distribution is the discrete analog of the exponential distribution for the continuous case, which is well-known in the community of mathematics and statistics (Johnson et al., 2005, p. 2010). The actual goal is to instruct readers from other backgrounds on the fact a geometric distribution for some random variable x can be expressed as an exponential

³The point is assuming that there are impossible orders; whether acceptable orders vary in degree of acceptability is irrelevant for this point.

function in a literal sense, that is involving an expression of the form

$$p(x) = \dots e^{f(x)}, \quad (\text{A1})$$

where $f(x)$ is some function of x . One of the targets of this appendix is the use of the term exponential distribution or the assumption of an expression of the form of equation A1 both for continuous and discrete variables (Clauset et al., 2009; Ferrer-i-Cancho, 2004; Newman, 2005) neglecting that, in the discrete case, that has a precise equivalent that is the geometric distribution.

The customary definition of an exponential distribution for a discrete random variable, e.g., frequency rank (Equation 2) and the definition of a geometric distribution (Equation 6) are indeed equivalent. To see it, notice that

$$\begin{aligned} p(r) &= c(1-q)^{r-1} \\ &= ce^{(r-1)\log(1-q)} \\ &= c'e^{-\beta r}, \end{aligned}$$

where $c' = c/(1-q)$ and $\beta = -\log(1-q)$.

Appendix B. The right-truncated geometric distribution

Our 2-parameter right-truncated geometric distribution is defined on $r = 1, 2, \dots, R$. Wimmer and Altmann (1999) present a similar but not equivalent 2-parameter right-truncated geometric distribution that is defined on $r = 0, 1, 2, \dots, R$. Our 2-parameter right-truncated geometric distribution is obtained when the normalization factor in Equation 6 is $c = 1/S(1, R)$, where

$$S(1, R) = \sum_{r=1}^R (1-q)^{r-1}.$$

A compact expression for $S(1, R)$ is easy to obtain. By the self-similarity property of the geometric series,

$$(1-q)S(1, R) = S(1, R) - 1 + (1-q)^R.$$

After some simple algebraic manipulations, one obtains

$$S(1, R) = \frac{1 - (1-q)^R}{q}$$

and then

$$c = \frac{q}{1 - (1-q)^R}.$$

It is easy to check that

$$\lim_{R \rightarrow \infty} c = q,$$

which is the normalization factor of the untruncated geometric distribution.

Appendix C. The value of R that maximizes log-likelihood

We aim to show that maximum $\mathcal{L}$ requires $R = r_{max}$ for the Zeta 2 model and the Geometric 2 model. First, we show that maximum $\mathcal{L}$ requires $R \geq r_{max}$. As these models are such that $p(r) = 0$ for $r > R$, setting $R < r_{max}$ implies that the likelihood of the model is $L = 0$ (recall Eq. 9) and the log-likelihood $\mathcal{L}$ goes to $-\infty$. Second, we show that the log-likelihood functions of these models are monotonically decreasing functions of R when the other parameter (α or q) is constant and thus maximum $\mathcal{L}$ requires $R = r_{max}$. Let us assume that α and q are constant. For Zeta 2, we revisit previous arguments (Baixeries et al., 2013).

The only summand in Eq. 11 that depends on R is $-F_0 \log H(R, \alpha)$. The recursive definition

$$H(R, \alpha) = \begin{cases} 1 & \text{if } R = 1 \\ H(R - 1, \alpha) + R^{-\alpha} & \text{if } R > 1 \end{cases}$$

clearly shows that $H(R, \alpha)$ is a monotonically increasing function of R (when α is fixed) and that $-F_0 \log H(R, \alpha) < 0$ since $F_0 > 0$ and $H(R, \alpha) \geq 1$. Therefore, $\mathcal{L}$ is a monotonically decreasing function of R (for constant α). For Geometric 2, we recall the assumption $q \in (0, 1)$ and note that the 1st derivative of $\mathcal{L}$ (Eq. 12) with respect to R is

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial R} &= -F_0 \frac{\partial \log(1 - (1 - q)^R)}{\partial R} \\ &= F_0 \frac{(1 - q)^R \log(1 - q)}{1 - (1 - q)^R}. \end{aligned}$$

It is easy to see that $\partial \mathcal{L} / \partial R < 0$ because all terms in the expression are strictly positive except $\log(1 - q) < 0$.

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